
SSM Optimization

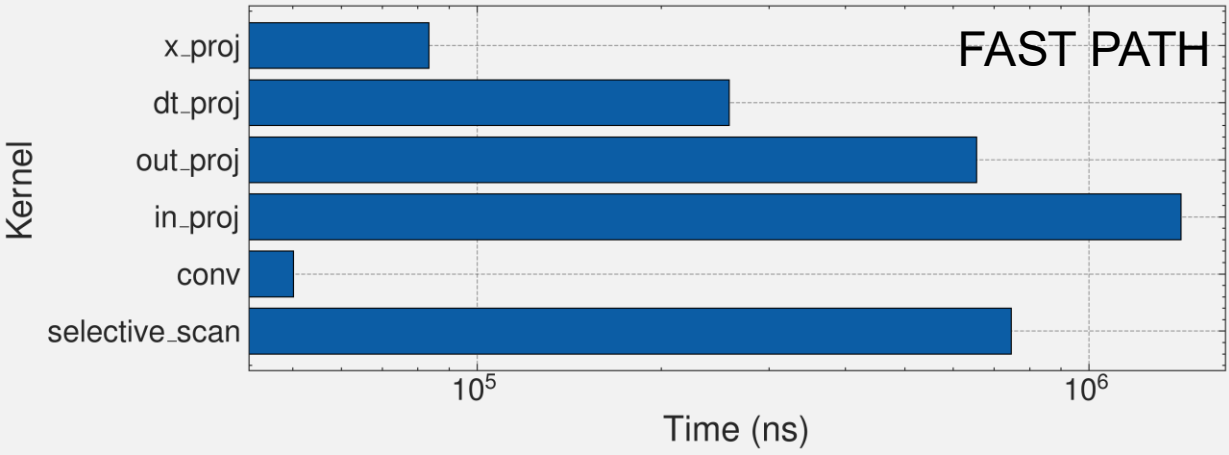
Weekly Meeting

03/07/2026

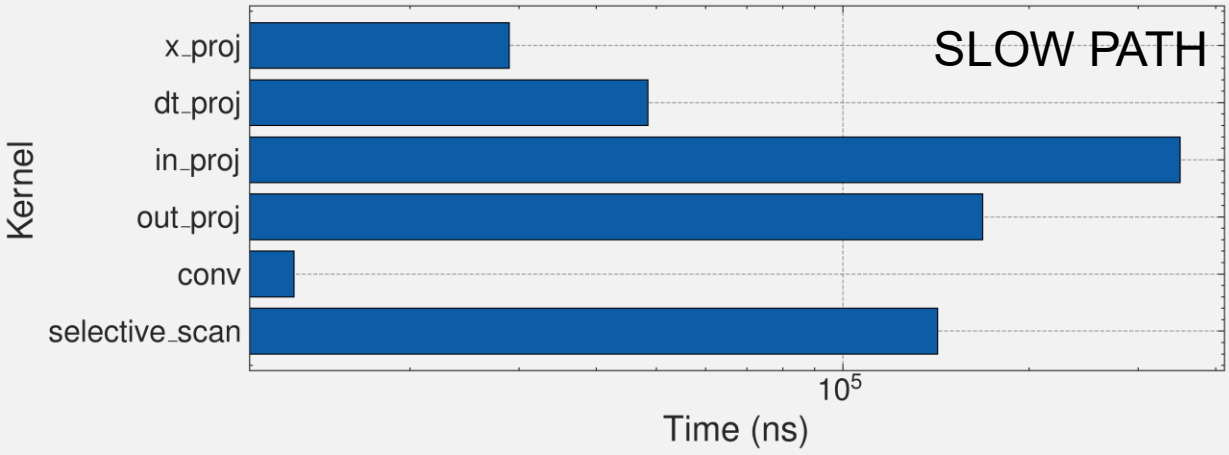
Profiling

- I had done a roofline model on paper, for the slow Pytorch forward implementation (recurrent), and the fast kernel
- I wanted to check the alignment with the actual arithmetic intensities prior to delving into the state compression experiments
 - Worked on profiling of the various blocks

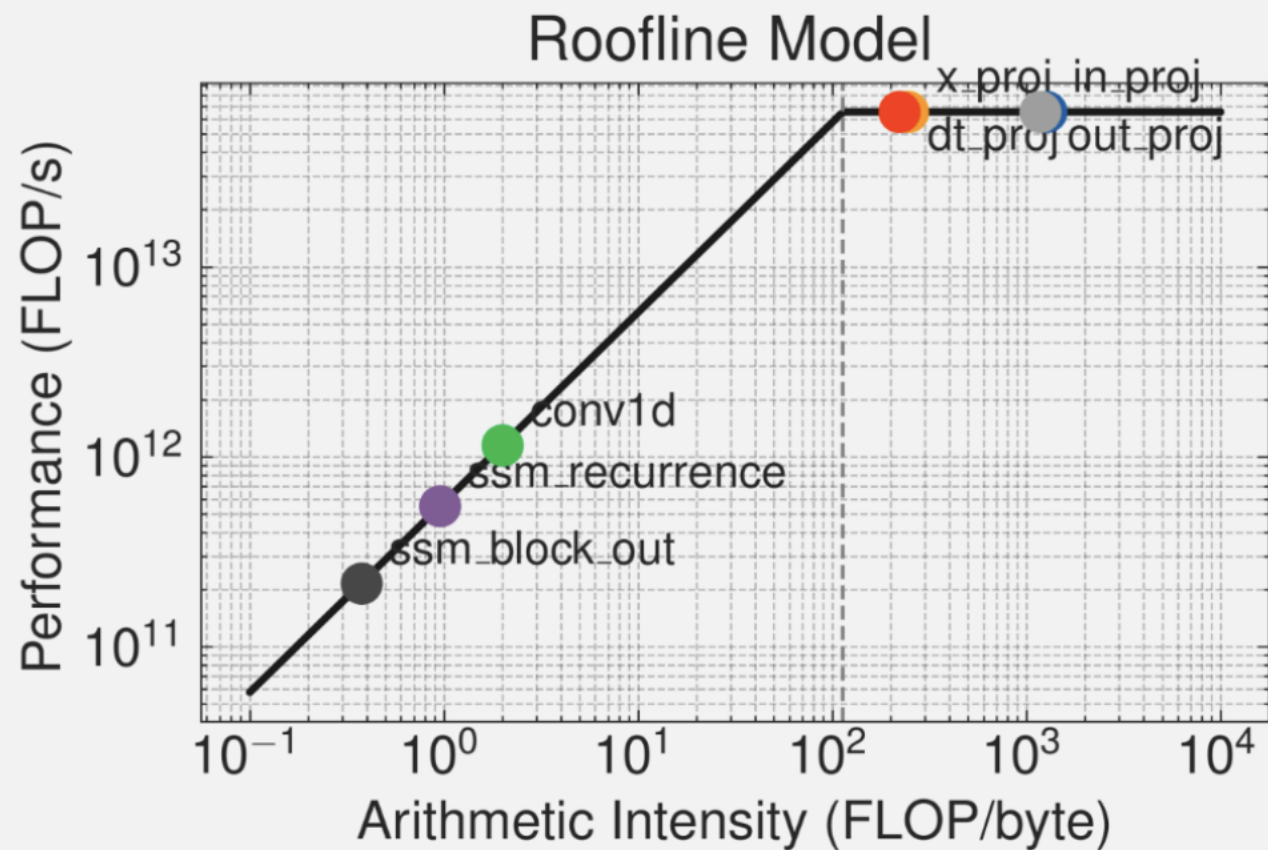
Execution times



Batch size 1, Sequence length 128

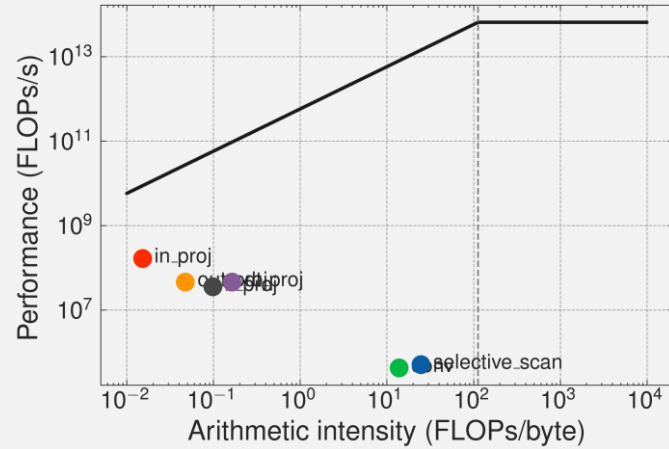


Old roofline

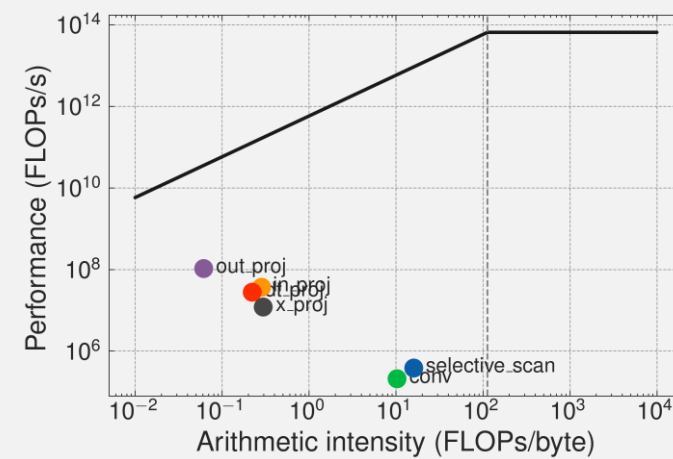


Roofline (profiled)

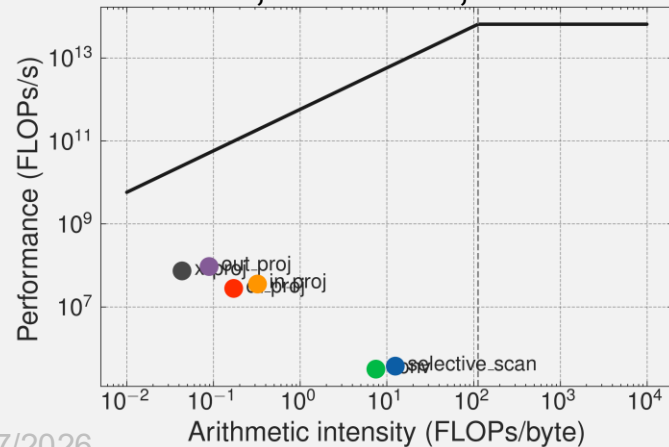
BS 1, SL 128, SLOW



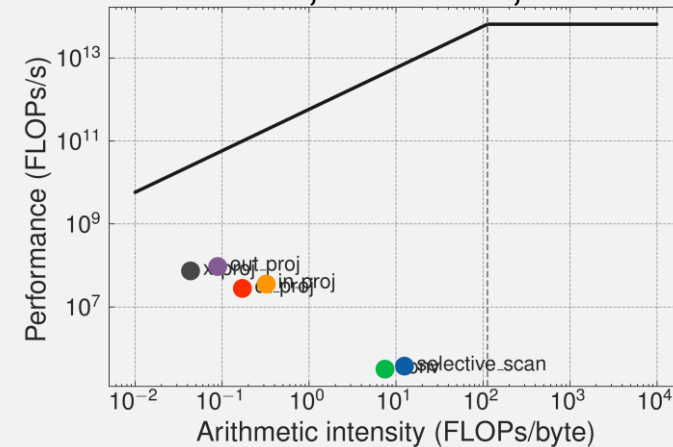
BS 32, SL 128, SLOW



BS 1, SL 1024, SLOW

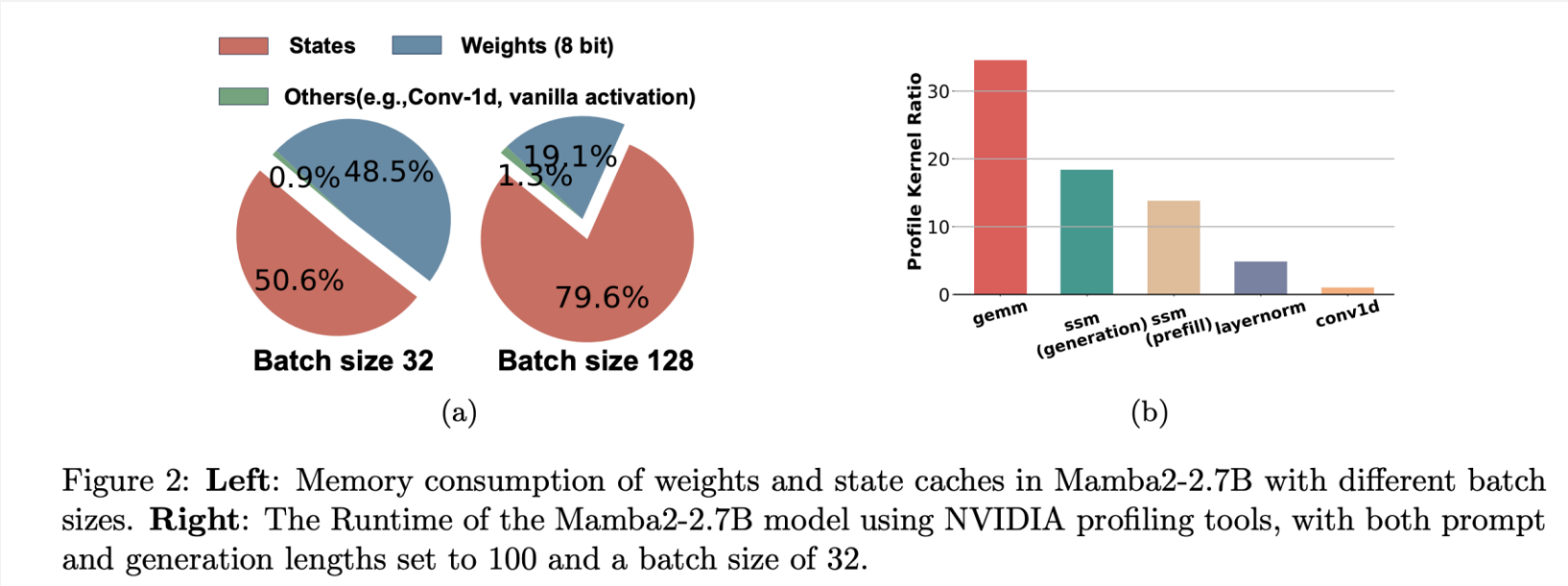


BS 32, SL 1024, SLOW



State impact

	Falcon Mamba 7B	
	BS = 32	BS = 128
Parameters	7,272,665,088	7,272,665,088
State values	268,435,456	1,073,741,824



Next steps

- Continue work on profiling
- Activations quantization